Software Technology 01

Course Introduction

What is Software Technology?

Midjourney “thinks”:

What is Software Technology?

ChatGPT “thinks”:

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What is software technology?

Software technology refers to the collection of tools, techniques, and processes used to create, design, develop, test, maintain, and deploy software applications. It includes various programming languages, frameworks, libraries, development methodologies, project management techniques, testing and debugging tools, and other software-related technologies.

Software technology plays a critical role in the creation of software applications, as it provides the means to design and develop complex and reliable software systems. The goal of software technology is to enable developers to create software that meets user requirements, is easy to use, and can be maintained and updated over time.

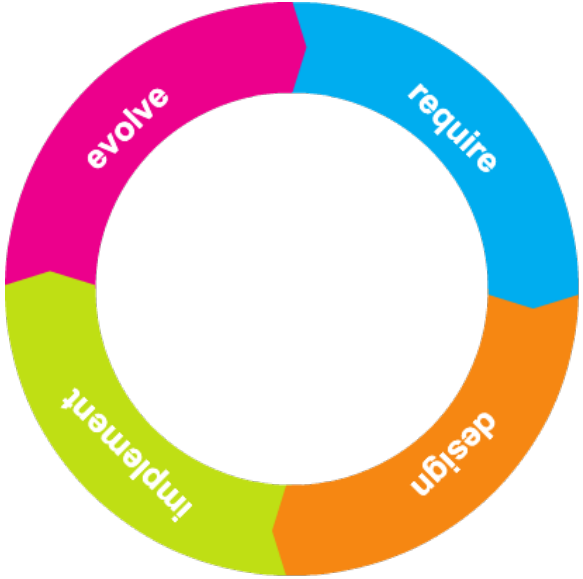
Some of the key areas of software technology include web development, mobile app development, database management, cloud computing, artificial intelligence, machine learning, and cybersecurity. These areas are constantly evolving, and new technologies and tools are being developed to improve the efficiency, security, and reliability of software systems.

What is Software Technology?

- Course? Faculty? Program? → too broad
- SW Engineering? SW Development?
- „**Software development** is the process of computer *programming*, *documenting*, *testing*, and *bug fixing* involved in creating and *maintaining* applications and frameworks resulting in a **software product**.”

- Ultimate goal: Software Product
- But how?

Software Development



Software Development Detailed

Contradictions

- Do we **optimize** for
  - Release date?
  - Unique product which is hard to copy?
  - Quality?
  - User Experience? (UX)
  - Developer Experience?
  - Easy Maintenance?
  - Easy Deployment?
  - Easy Usage and Compatibility?
  - Vendor Lock-In? :)
  - Marketable product?
    (what to write on the box, number of features)
  - Platform independence?
  - Performance?
  - Cost?
  - Least amount of useless work?

Software Development can be ~~Good~~ **Fit for the job**

SW Development History

- Software Development
  - Processes
  - Methodologies
  - Philosophies →
  - & Trends

- Acceptance test–driven development
- After the Software Wars
- Agile Manifesto
- Agile software development
- Pull-based agile coaching
- Behavior-driven development
- Best practice
- The Cathedral and the Bazaar
- Comment programming
- Composition filters
- Cowboy coding
- Design-driven development
- Domain-driven design
- Extreme programming
- Formal methods
- Homesteading the Noosphere
- Integration competency center
- Iterative and incremental development
- Kanban (development)
- KISS principle
- Lean integration
- Lean software development
- Lightweight methodology
- The Magic Cauldron (essay)
- Mayo-Smith pyramid
- Micro-innovation
- Minimalism (computing)
- Open/closed principle
- Planning poker

Why this course?

- Bridge the gap between university and work experience
- Learn and try software technology in a “real world setup”
- MSc vs. BSc:
  - Innovation (academic, technical)
  - Organization & leadership skills

Output

- Roles we are shooting at:
  - Manager
  - Architect
  - Lead Developer
  - Knowledgeable Specialist
  - ...anyone with Product Mindset

Structure of Course

- Assignment: The Project
- Lecture
  - Semester 1st part: The Process
    - Software Development Models, Methodologies and Tools
  - Semester 2nd part: The Architecture
    - Architectural Improvements, Design Patterns

Technical details: See in Canvas

- Contacts (Team-level communication)
- Requirements
- Lectures, timing
- Practice, consultation: defined by the mentor

The Project Game

- Suggested (now trending) fields
  - Make the World a Better Place!
  - Motivation
  - Innovation
  - Technical Expertise
- Future (aka Take it Seriously!)
  - Sketched Idea & Business Model
  - Sustainable Progress, Maintainable Product
  - The Team (carefully assess and select!)
  - Startup, Investors, Success

How to qualify

- Goal: MVP = Minimum Viable Product
- Teams of 5 (or 4, if needed)
- Successful project execution
  - Documented (concisely) –
    - Software Design (UX) – towards user
    - Product Architecture – towards maintainer
    - Execution Planning and Chosen Methodology
    - Development Process and Delivery Chain
  - Working and Maintainable –
    - Software Product
    - Test and Build System
    - Automated Release

Requirements

- Single mark for all work during whole semester
- Evaluation of Teams
  - Balanced teamwork effort
- Everything is **Team Responsibility**
- Everything should be put into **VCS** (version control) (we will check individual commits)
- Participate on Weekly Consultation

Final Star Distribution

- Equal Contribution
- Fair Distribution
- Min. 1 star
- Min 50%

+ Final Test Quiz

Name:	Range:	
25(T5) or 20(T4)	100 %	to 87.0%
24(T5) or 19(T4)	< 87.0 %	to 85.0%
23(T5) or 19(T4)	< 85.0 %	to 84.0%
23(T5) or 18(T4)	< 84.0 %	to 83.0%
22(T5) or 18(T4)	< 83.0 %	to 81.0%
22(T5) or 17(T4)	< 81.0 %	to 80.0%
21(T5) or 17(T4)	< 80.0 %	to 78.0%
20(T5) or 16(T4)	< 78.0 %	to 75.0%
19(T5) or 15(T4)	< 75.0 %	to 73.0%
18(T5) or 15(T4)	< 73.0 %	to 72.0%
18(T5) or 14(T4)	< 72.0 %	to 70.0%
17(T5) or 14(T4)	< 70.0 %	to 69.0%
17(T5) or 13(T4)	< 69.0 %	to 68.0%
16(T5) or 13(T4)	< 68.0 %	to 66.0%
16(T5) or 12(T4)	< 66.0 %	to 65.0%
15(T5) or 12(T4)	< 65.0 %	to 63.0%
14(T5) or 11(T4)	< 63.0 %	to 60.0%
13(T5) or 10(T4)	< 60.0 %	to 58.0%
12(T5) or 10(T4)	< 58.0 %	to 57.0%
12(T5) or 9(T4)	< 57.0 %	to 55.0%
11(T5) or 9(T4)	< 55.0 %	to 54.0%
11(T5) or 8(T4)	< 54.0 %	to 53.0%
10(T5) or 8(T4)	< 53.0 %	to 50.0%
Team Failed	< 50.0 %	to 0.0%

Deadlines – see Canvas

Team Facilitation

1) Startup Idea Pool
2) Mentors select best ideas (top 20%)
3) Mentors are assigned to ideas
4) Idea owners put together Team
    1) Set Canvas Profile Picture!!!
    2) Use Discussions (Forum) to find members or teams!
5) Consultation Starts with Mentor, Idea Elaborated
6) See tasks and deadlines in Canvas!

Be Open!

- Share Your Idea, Get Feedback
- Make Profile Picture and CV on your technical skills for Idea Owners!
- What Makes a Good Team
  - Members Selected for Idea by – **GOOD**
    - Motivation
    - Technical Skills
  - Idea Selected for Members by – **EXTREMELY DANGEROUS!!!**
    - Friendship
    - Speaking the same language
    - Drinking Buddies
- Work Openly https://opensource.guide/